

Sandrina Mosley Test Strategy

CUCM to Microsoft Teams Voice - Verification & Validation

Document Purpose

This test strategy defines how a CUCM-to-Microsoft Teams Voice migration will be verified before, during, and after implementation. It provides a repeatable approach for functional testing, integration testing, network and voice-quality validation, security checks, resiliency, user acceptance, regression testing, and rollback validation.

1. Test Objectives

- Verify that Teams Voice services meet documented business and technical requirements.
- Confirm end-to-end call flows between Teams, carrier services, PSTN, and remaining CUCM components during coexistence.
- Validate emergency calling behavior and location accuracy.
- Verify voice quality, network readiness, resiliency, and failover behavior.
- Confirm that operational support and user workflows are ready for production.

2. Test Environment

- Microsoft Teams tenant with production-equivalent voice configuration.
- Configured user accounts, phone numbers, resource accounts, auto attendants, call queues, and policies.
- SBC/carrier connectivity where Direct Routing or equivalent architecture is used.
- Representative endpoints: Teams desktop, mobile, web, certified desk phones, headsets, meeting devices, and shared devices as applicable.
- CUCM coexistence environment and documented test numbers.

3. Test Types

Test Type	Purpose	Example Coverage
Functional	Verify individual Teams Voice functions	Inbound/outbound, hold, transfer, voicemail, conferencing
Integration	Verify systems work together	CUCM, SBC, carrier, directory, emergency services
Network/QoS	Verify real-time media readiness	Latency, jitter, loss, QoS, bandwidth
Security	Verify secure operation	Admin roles, authentication, encryption, logging
Resiliency	Verify service continuity	SBC failover, WAN path, endpoint recovery
UAT	Verify business usability	User scenarios, call flows, support readiness
Regression	Verify unchanged services remain functional	Previously validated call paths and policies
Rollback	Verify recovery path	Return selected users/routes to CUCM

4. Test Preparation

- Trace each test to a business or technical requirement.

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- Create test users, test numbers, service accounts, and representative endpoints.
- Confirm test windows, contacts, evidence repositories, defect process, and approval workflow.
- Define entry and exit criteria before execution begins.

5. Functional Test Cases

ID	Test Case	Expected Result
UC-001	Teams user sign-in and policy assignment	User signs in and receives intended voice configuration
UC-002	Outbound internal call	Call connects with correct audio and identity
UC-003	Outbound PSTN call	Call reaches intended destination with correct calling number
UC-004	Inbound PSTN call	Call reaches intended Teams user or service
UC-005	Hold and resume	Two-way audio resumes correctly
UC-006	Blind transfer	Transferred call reaches destination
UC-007	Consultative transfer	Consult and transfer complete successfully
UC-008	Voicemail	Unavailable call is routed and message is accessible
UC-009	Caller ID	Configured calling identity is presented correctly
UC-010	Auto attendant	Menu options route to intended destination
UC-011	Call queue	Queue distributes calls according to configuration
UC-012	After-hours routing	Calls follow schedule and holiday behavior
UC-013	Mobile client	User can place and receive authorized calls
UC-014	Desk phone	Certified device registers and supports required features

6. Integration Test Cases

ID	Test Case	Expected Result
INT-001	Teams-to-CUCM coexistence call	Call connects with expected signaling and media
INT-002	CUCM-to-Teams call	Call reaches Teams destination
INT-003	Teams-to-PSTN via carrier	Call completes with expected numbering and audio
INT-004	Inbound PSTN number routing	Number routes to intended user/service

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ID	Test Case	Expected Result
INT-005	Number normalization	Numbers are translated consistently
INT-006	SBC path validation	Signaling/media traverse intended interfaces
INT-007	Directory/user synchronization	User identity matches intended Teams object
INT-008	Emergency location mapping	Test emergency workflow returns correct location data

7. Network and Voice Quality Testing

- Validate packet loss, jitter, latency, bandwidth, DNS, NTP, routing, and QoS at representative sites.
- Place test calls during normal and peak periods where practical.
- Capture available call-quality telemetry and endpoint statistics.
- Investigate one-way audio, clipping, delay, failed media negotiation, or unexpected call setup times.
- Document baseline and post-migration observations by site.

8. Security Testing

- Verify administrative roles follow least-privilege principles.
- Verify authentication and account controls for privileged operations.
- Validate secure signaling/media configuration and certificate status where applicable.
- Confirm audit/logging requirements and escalation process.
- Verify that test and production data are appropriately separated.

9. Resiliency and Failover Testing

- Simulate loss of a primary network path and verify recovery behavior.
- Test SBC or carrier redundancy where implemented.
- Verify endpoint recovery after network interruption or client restart.
- Validate business continuity for critical numbers and call flows.
- Record recovery time observations and identify gaps.

10. User Acceptance Testing

- Select representatives from business units, locations, executives, service desk, and other important user groups.
- Execute real-world scenarios such as placing calls, receiving calls, transferring calls, voicemail, mobile use, and shared services.
- Capture user acceptance, comments, defects, and training needs.
- Require business owner approval before production expansion.

11. Defect Management

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Severity	Definition	Disposition
Critical	Core voice service unavailable or business-critical calling fails	Immediate triage; migration blocked until disposition
High	Major function fails with significant business impact	Correct and retest before affected wave
Medium	Important function impaired with workaround	Track and resolve within agreed release plan
Low	Minor issue with limited impact	Track for later resolution

12. Test Entry Criteria

- Target voice configuration is complete for the test scope.
- Required test users, numbers, endpoints, and services are available.
- Test cases and expected results are approved.
- Test support contacts and defect process are established.

13. Test Exit Criteria

- All planned critical and high-priority tests have passed or have documented approved dispositions.
- Emergency calling validation is complete for the migration scope.
- Critical call flows and integrations are functioning.
- User acceptance results are documented.
- Rollback validation is complete.
- Test evidence and sign-off package are ready.

14. Sample Test Execution Record

Field	Example
Test ID	UC-001
Test Description	Teams user sign-in and voice configuration
Preconditions	Licensed user; required policy assigned
Steps	Sign in; verify calling tab; place test call; receive test call
Expected Result	User is enabled and calls function normally
Actual Result	Pass
Evidence	Screenshots, call logs, test notes
Defect ID	N/A
Tester	Assigned Test Engineer
Date	To be completed during execution

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15. Migration Wave Validation

- Before wave: confirm user list, numbers, licensing, policies, emergency locations, and support coverage.
- During wave: validate representative inbound/outbound calls and priority services.
- After wave: monitor incidents, call quality, user feedback, and carrier/SBC status.
- Approve the wave only after documented exit criteria are met.

Portfolio Disclaimer

This document is a sample unified communications test strategy created for demonstration and portfolio purposes. It uses generalized business and technical scenarios and is not a reproduction of proprietary employer or customer documentation.